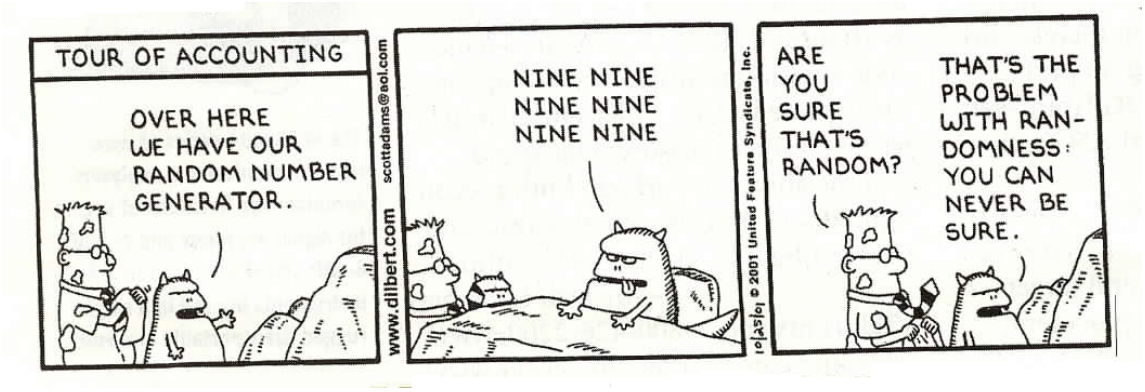


Homework #1 (due Oct. 23)

On your own (no help from Claude, other than for debugging):

- Modify the provided program "rnd_test.f" to test, at a minimum, three of the following different random number generators: the intrinsic function RND( ) (see program "example4.f"), the subroutine RANDM(X) shown in "example3.f", the subroutine RANDU described in the randu.pdf handout (copy it from that document, it is just a few lines), the RANDOM_NUMBER intrinsic function (see https://gcc.gnu.org/onlinedocs/gfortran/), or the more sophisticated R250 generator available from https://www.taygeta.com/random.html. Notice the very small (but perhaps significant) difference between RANDM(X) and RANDU. You can also transcribe other random number generating subroutines from the "Numerical Recipes" book at Crerar library, if you feel inclined, from papers posted in Canvas, or any other source. Vary the seed to see if this has any repercussions in the quality of the random numbers generated (does IX have to be a 5-digit odd number for RANDM(X), or is this mere superstition?). The suggested method is to check for statistically significant deviations from the mean in histogram bins into which the random numbers have been sorted, exploring this down to some decimal place -try to go as deep as possible-. Plot your histograms and express your conclusions about the respective qualities of these routines. Experiment with the numerical precision of the variable holding the random number (see DOUBLE PRECISION statement).

Vibe-coding your way into oblivion:

- Try the same using other methods discussed in class, as many as you can: correlations between triplets, frequency of appearance of 0-9 integers at a certain (deep) decimal position, "maximum spacing" method in provided paper by J. Heinrich, or anything else your robot overlord might suggest. **Do not cease until you have anomalies in random # generation to report** (it is fine to let Claude generate this second part of the report).